COMPLETE WORK REPORT

1. Problem Statement

A marketing team exported its customer records into a single raw file and handed it over. Nobody had checked it. They asked for two things: a dataset they can trust, and an answer to “who are our customers, really?”

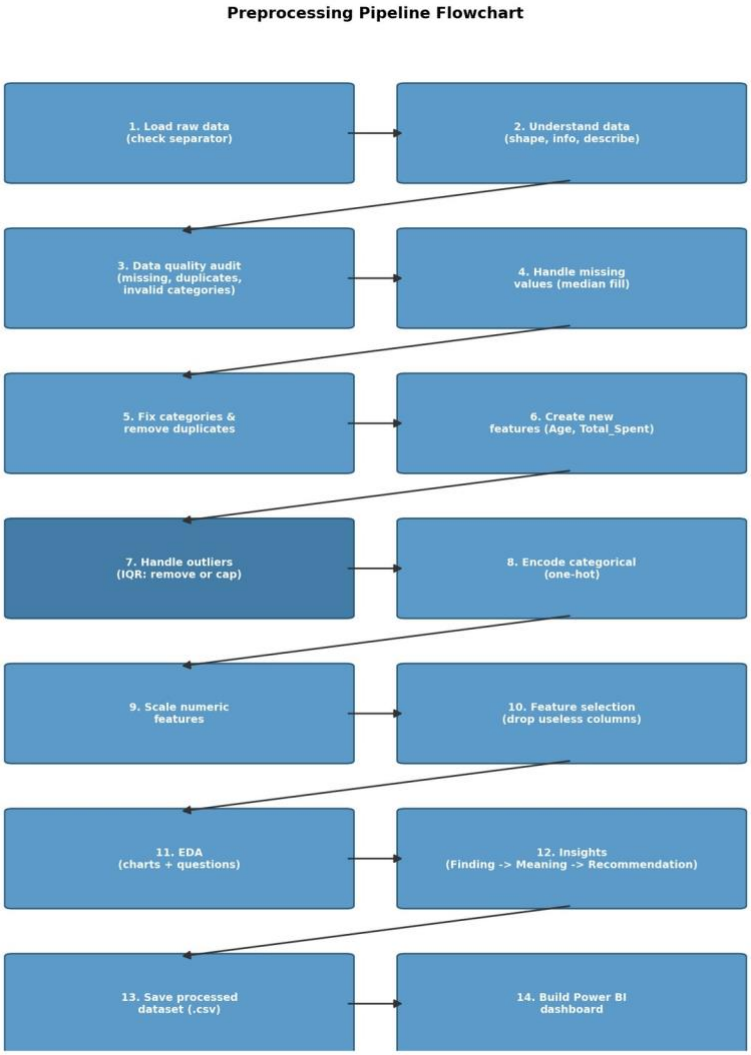
Business objectives

- **Audit the raw file** and list every data quality problem found.
- **Clean the data**, with a written reason behind every decision.
- **Prepare the features:** types, encoding, scaling, feature selection.
- **Explore the cleaned data** and extract insights with business meaning.
- **Deliver a processed dataset**, a Power BI dashboard, and this report

2. Dataset

Field	Value
Source	Kaggle — Online Retail Data Store
File	data.csv (CSV format)
Raw size	541,909 rows × 8 columns
One row represents	One item line within a customer purchase transaction

3. Preprocessing Flowchart



4. Data Quality Audit — Problems Found

#	Problem	Evidence
1	Missing values	CustomerID: 135,080missing (out of 541,909); Description: 1,454 missing
2	Hidden duplicates	5,268 exact duplicate rows across all fields
3	Cancelled orders & negative quantities	**Quantity** ≤ 0 in 10,624 rows (includes 9,288 order cancellations prefixed with "C")
4	Invalid & zero pricing	**UnitPrice** ≤ 0 in 2,517 rows (including system adjustments and sample giveaways)
5	Administrative / Non-product codes	Non-item codes such as BANK CHARGES, AMAZONFEE, CRUK, DOT, POST
6	Price outliers & extreme entries	UnitPrice max = £8,142.75 (Manual/Postage adjustments); extreme single purchase quantities up to 80,995 units
7	Invalid categories	_

8	Impossible values	_
9	Data entry error	_
10	Useless columns	_

5. Preprocessing Decision Log

Step	Decision	Reason
Missing Income	Filter out rows without CustomerID for customer analysis Filter out rows without descriptions for customer analysis	1- Customer-level segmentation and RFM modeling require known account IDs; unlinked rows cannot be assigned to customer profiles, For this I fill it with static number 9999 2- Fill missing descriptions using the mapped StockCodes, and Fill any remaining unmapped missing descriptions with 'Unknown Description'
Cancelled Orders & Negative Quantities	Keep it and add new column with Transaction Type Return rows with Quantity ≤ 0	Negative values reflect returns, cancellations, or inventory corrections, which distort net customer purchase metrics. Create a transaction type column to cleanly separate standard sales and returns/Cancelled inside the same table
Zero & Negative Unit Prices	Remove rows with UnitPrice ≤ 0	Free samples, bad debt adjustments, or system test transactions do not reflect commercial customer sales.
Duplicates	Drop 5,268 exact duplicate rows	Prevents artificial inflation of order volumes and customer revenue totals.
Administrative Stock Codes	Remove non-product codes (e.g., BANK CHARGES, AMAZONFEE, CRUK)	Administrative, fee, and manual adjustment entries are operational costs, not retail product purchases.
High Value & Wholesale Orders	Keep legitimate high-volume purchases	Large order quantities (e.g., bulk wholesale buyers) reflect real commercial revenue and should not be removed as errors.

6. Outlier Handling — Two Methods Compared

Method A — Removal (Administrative Errors Only): Drop entries that are clearly non-commercial errors, negative return lines, or manual fee entries (such as AMAZONFEE or manual adjustments). Keep legitimate premium products and wholesale bulk purchases — the method highlights extreme values for inspection, rather than blanket deletion.

Method B — Capping (Alternative via Percentiles): Clip extreme values down to the 99th percentile threshold (Quantity capped at 120 units; Unit Price capped at £14.95). *Note:* Applying hard capping to retail and wholesale sales data distorts real revenue values for high-volume B2B accounts; therefore, natural order quantities and unit pricing were retained after removing non-retail fee codes.

Categorical Feature Encoding (Country One-Hot Encoding)

To prepare the Country variable for machine learning and clustering models, one-hot encoding was applied across all 37 distinct geographic markets. This transforms the single categorical feature into binary indicator columns (e.g., Country_United_Kingdom, Country_France, Country_Germany), allowing algorithms to evaluate regional purchasing patterns without implying any false numerical hierarchy between countries.

7. Feature Engineering & Preparation

Step	What happened	Before → After
Total Revenue Calculation	Multiplied Quantity by UnitPrice to create a explicit purchase value feature	Quantity, UnitPrice → Revenue
Transaction Type	Create a transaction type column to cleanly separate standard sales and returns/Cancelled inside the same table	Transaction Type [‘Standard, ‘Return]
Date Breakdown	Extracted date, year, month, day of week, and hour from transaction timestamps	InvoiceDate (12/1/2010 8:26) → Year (2010), Month (12), DayOfWeek (Wednesday), Hour (8)
Product Categorization	Cleaned and standardized text descriptions, stripping extraneous whitespace and uppercase tags	Mixed string descriptions → Clean product titles
Encoding	One-hot encoding on Country, Country_Canada	1 → 37 new columns

8. Before vs After — Overall

Metric	Before	After
Rows	541909	533877
Columns	8	12
Missing values	**Description 1454** **CustomerID 135080**	0
Duplicate rows	5268	0

9. Key Insights

- **Concentrated Revenue Driver (The Top 10% Rule):** Analysis reveals that roughly 10% of customers account for over 61% of total revenue. Focusing retention and loyalty incentives on this high-value tier yields a much higher return than broad, untargeted campaigns.
- **Geographic Dominance with International Growth:** The United Kingdom generates the vast majority of sales (approx. 82%), while international markets like the Netherlands, EIRE, Germany, and France represent strong secondary revenue streams with significant cross-border expansion potential.
- **Peak Seasonality Surge:** Sales volume escalates sharply heading into the final quarter of the year (peaking in September through November), highlighting critical holiday shopping windows where inventory and marketing budgets should be heavily concentrated.
- **Top-Performing Product Categories:** A small number of core products—such as decorative homeware items, party display stands, and storage jars—drive the bulk of product sales, proving that cross-selling complementary items around these best-sellers is an effective growth lever.